

## 2.7 File Ownership And Permissions

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We'll never tire of telling you that many of the advantages of Linux are a result of its springing from a strong multi-user operating system. In a multi-user system, it is important to ensure that the files of one user cannot be accessed by another user unless the latter is permitted to access those files. This gives the idea of file ownership. The user who creates a file is, by default, the owner of that file. He can do whatever he wishes to do to his file, such as editing and deleting. A user group can also own a file, if a member of the group has created the file. The user group will have the same permissions as that of the file creator. This is an important feature, because there can be instances where a group of users are working on a project and files created by one user have to be accessible and editable by others in the group.

Besides the owner of the file and his user group, other users can be granted privileges that will enable them to work on the files created by the owner.

There are three basic file permissions in Linux: Read Permission, Write Permission and Execute Permission. The read and write permissions behave slightly different on a file and on a directory. The read permission on a file enables any user who is granted this permission to open and view the contents of the file, but not make changes to it. On a directory, however, only the contents of the directory—the file listing—can be viewed. A file contained in the directory cannot be opened unless a read permission is given to that file.

On a file, the write permission allows you to write into the file, that is, make changes to the file. But on a directory, this permission enables you to add, remove or rename files in the directory. Hence, a write permission on a file will allow you to make changes to the file, but deleting the file or renaming it can be done only if the directory allows you to do that.

The execute permission, in case of a regular file, means you can execute the file as a program or a shell script. However, on a directory, the execute permission allows you to access files in the direc-